START HACK

Chat with Your Test Data

Deep Dive — Hackathon 2026

"Turning complex test data into simple conversations"

Zwick Roell

Agenda

1. **What is Material testing?**

2. **Why We Are Here?**
   Problem
   Vision
   What Good Looks like?

3. **Tech & Data**

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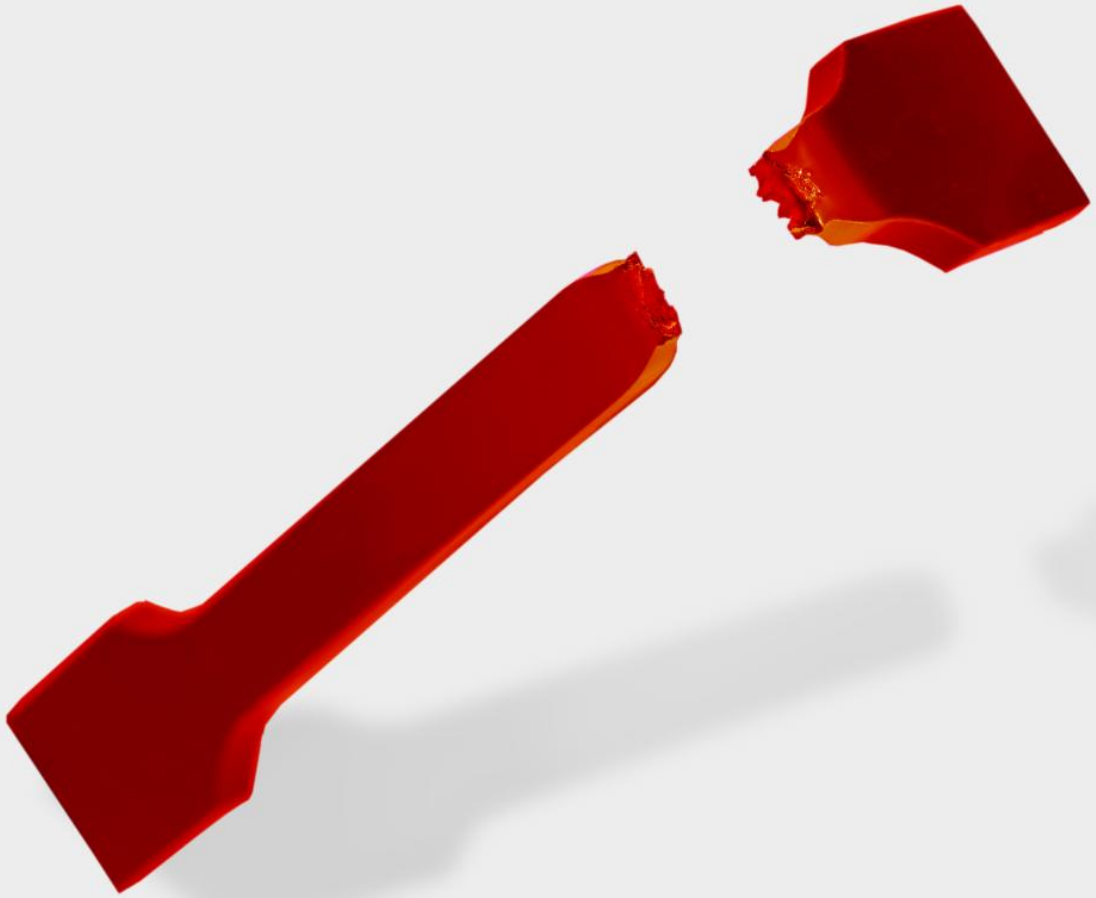

What is Material Testing?

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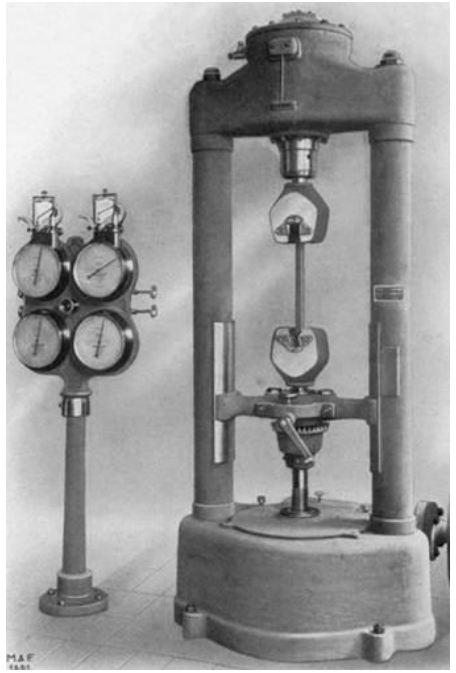
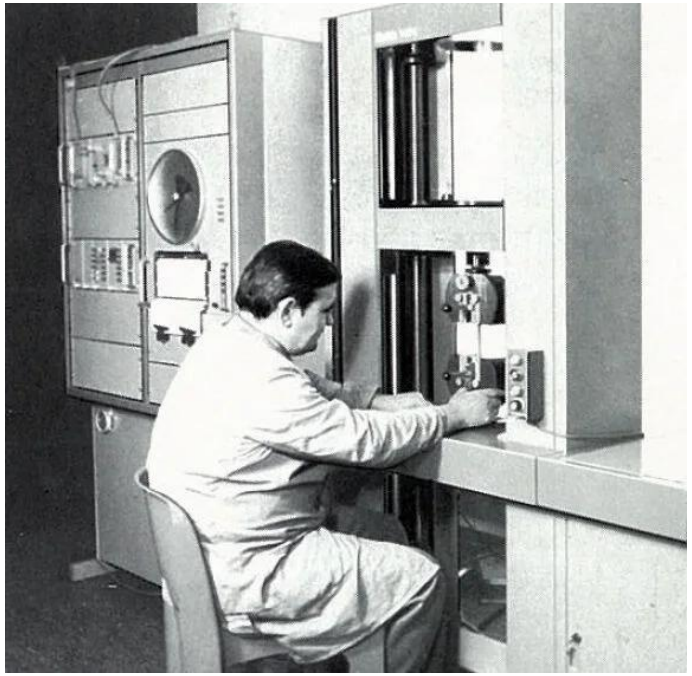
The Evolution of Measurement in Quality and Testing

Dynamometer, an instrument for measuring mechanical force or power, designed by the French civil engineer Edme Regnier (1751-1825).
Engraving from Encyclopaedia Londinensis (London, 1803).



Tensile testing machine for metals specimen manufactured by Mohr & Federhaff (1880s)



1974 Picture of the Zwick 1400 Series

The “present”

Material testing is fundamentally used to characterize material and product properties and functions

Research & Development

New Materials

Evaluate material properties during material development

New Product Innovation

Measure function and performance under defined parameters

Production

Quality Control

Secure constant quality during (mass) production

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What Can Be Tested?

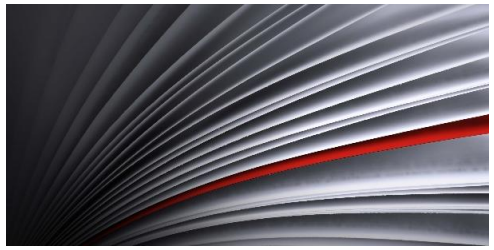
Metal

Plastics

Composites

Automotive

Medical Technology

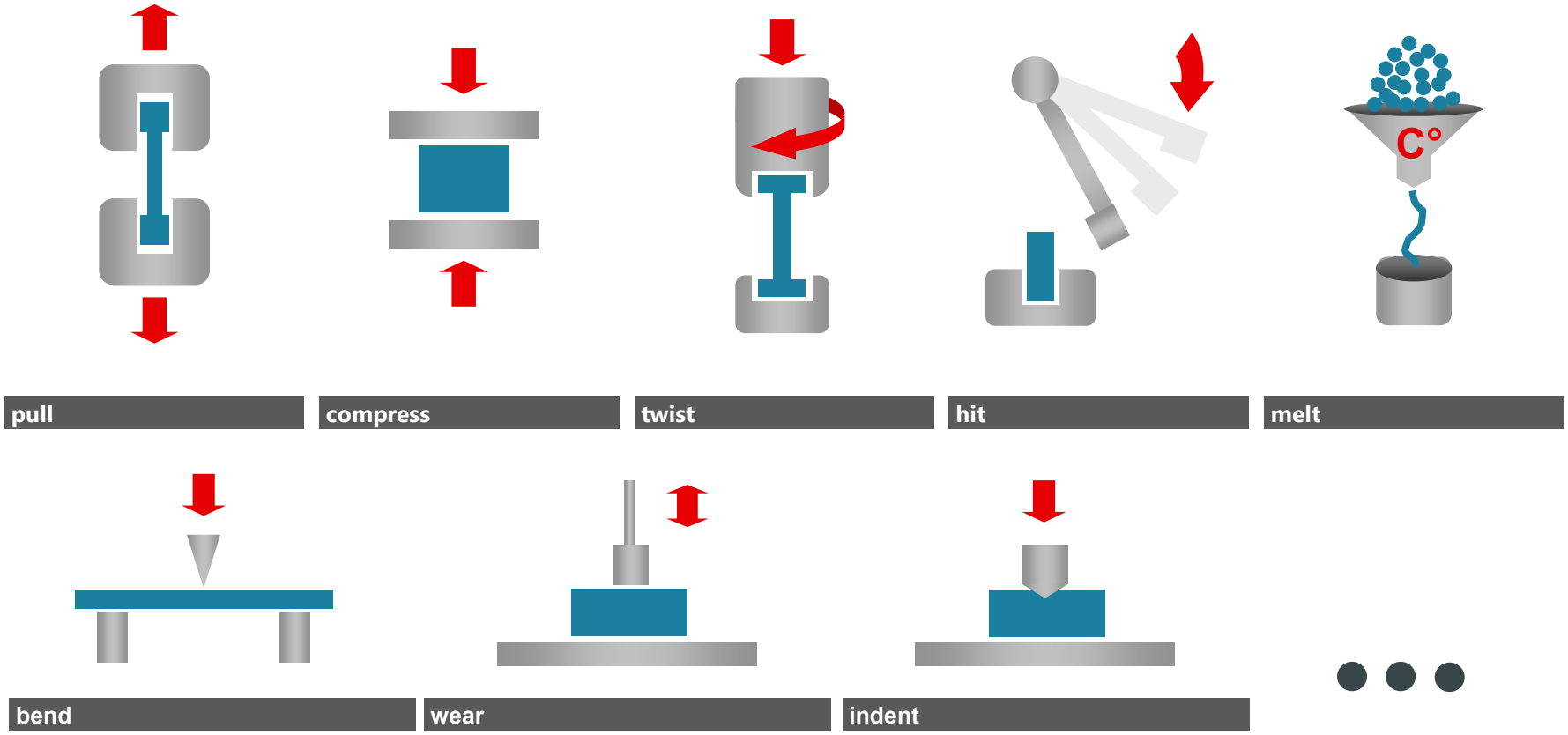


Paper & Cardboard

Textile

Food & Packaging

There are multiple ways and methods established over the years to characterize material properties. Just to show a few…



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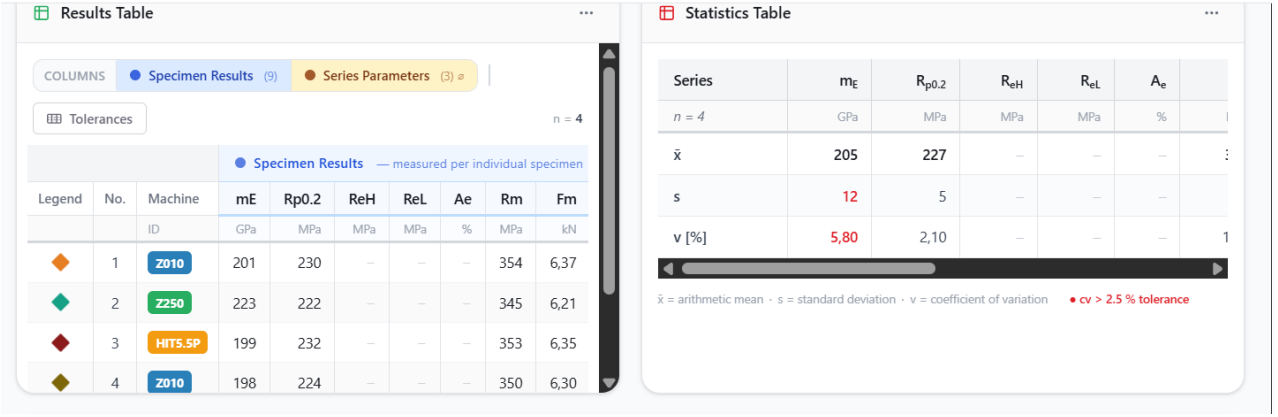
What testing Data Looks Like (No Engineering Needed)

Stress–Strain Curve



Legend	No.	Machine ID	mE GPa	Rp0.2 MPa	ReH MPa	ReL MPa	Ae %	Rm MPa	Fm kN
	1	Z010	201	230	–	–	–	354	6,37
	2	Z250	223	222	–	–	–	345	6,21
	3	HIT5.5P	199	232	–	–	–	353	6,35
	4	Z010	198	224	–	–	–	350	6,30

Series	m_E	$R_{p0.2}$	R_{eH}	R_{eL}	A_e
n = 4	GPa	MPa	MPa	MPa	%
x̄	205	227	–	–	–
s	12	5	–	–	–
v [%]	5,80	2,10	–	–	–

Results Table + Statistics Table

Histogram

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Snippets of testing data

How Reports Looks Today

Zwick Roell 29.01.21

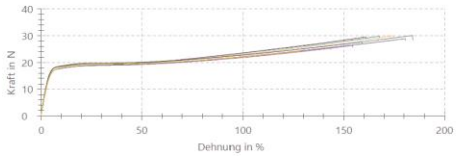
Prüfprotokoll

Kunde :
Auftrags-Nr. : OPP 233149
Versuch : Zugversuch
Prüfnorm : in Anlehnung an ISO 10555-1
Werkstoff : Blauer Schlauch mit OD Ø 1,25mm
Maschinendaten : Probenhalter: Schraub-PH Typ 8131 (Fmax 200N)
bzw. Pneumatik-PH Typ 8195 (Fmax 200N)
Backeneinsätze: Vulkollan glatt
Wegaufnehmer: Traverse
Kraftaufnehmer: Fmax 200N
Prüfer : ct-Sandra Klaiber
Vorkraft : 0,1 N
Prüfgeschwindigkeit : 2000 mm/min
Einspannlänge bei Startposition : 100 mm

Prüfergebnisse:

Legende	Nr	Probenhalter	Fmax N	εB %
	1.1	Schraub-PH Typ 8131	30,0	175
	1.2		29,8	168
	1.3		29,5	161
	1.4		25,4	144
	1.5		26,3	155
	2.1	Pneumatik-PH Typ 8195	28,9	181
	2.2		28,9	181
	2.3		27,6	159
	2.4		30,1	184
	2.5		24,1	128

Seriengrafik:



Zugversuch an Kunststoffschläuchen_blauer Schlauch mit OD Ø 1,25mm_OPP233149.zs2

Zwick Roell 29.01.21

Statistik:

Schraub-PH Typ 8131 n = 5	Fmax N	εB %
x̄	28,2	160
s	2,19	12,0
V [%]	7,76	7,49

Pneumatik-PH Typ 8195 n = 5	Fmax N	εB %
x̄	27,9	167
s	2,30	23,7
V [%]	8,26	14,25

Static PDF or Excel

Per test or test series, per method

Zwick Roell Print date: 02.02.21

Test report

Customer : -redacted-
Job no. : OPP.228903
Test standard : DIN EN ISO 6892-2 on 350°C
Material : Metallic arroy
Specimen type : M16 threaded head Ø 10mm
Tester : ct-Sam Liu
Machine data : Extensometer: LaserXtens (TZ)
Ceramic clip-on

Test speeds : Method B
Pre-load : 5 MPa
Speed, Young's modulus : 5 MPa/s
Speed, yield point : 5 MPa/s
Test speed : 0,09 1/min

Test results:

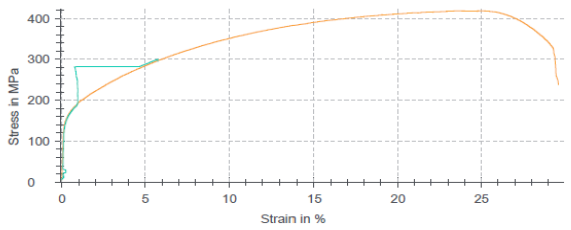
Legend	No.	mE GPa	Rp0.2 MPa	Rm MPa	Fm kN	Ag %	A50mm %	At %	d0 mm	S0 mm²	Temperature
	1	113	160	418	32,91	24,60	29,4	29,6	10,007	78,65	350°C
	2	140	158	301	23,65	5,47	-	-	10,005	78,62	350°C

Extensometer

Legend	No.	
	1	LaserXtens (TZ)
	2	Ceramic clip-on

Zwick Roell Print date: 02.02.21

Series graph:



Statistics:

Series n = 2	mE GPa	Rp0.2 MPa	Rm MPa	Fm kN	Ag %	A50mm %	At %	d0 mm	S0 mm²
x̄	127	159	360	28,28	15,04	29,4	29,6	10,006	78,63
s	19	2	83	6,55	13,53	-	-	0,001	0,02
□ [%]	15,26	1,19	23,13	23,16	89,95	-	-	0,01	0,03

Why We Are Here?

"Turning complex test data into simple conversations"

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Why We Are Here?

*Materials testing generates **enormous volumes of complex data** — and engineers spend too much time hunting for insights buried inside it. Your challenge: build the first prototype of **Chat with Your Test Data***

🧠 Real Problem

A genuine, industry-scale challenge and not a toy exercise

🚀 Your Mission

Prototype a conversational interface for test data

✨ The Twist

We show you *what*, not *how* — creativity is the point

> **"This is your chance to rethink how an entire industry interacts with its data"**

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The Data Is There. The Insights Are Locked Away

Fragmented Data

Data lives across testing methods, industry standards, materials and machines. Ther is not necessarily a natural linkage

Static Reports

Today the way is to read long, unnavigable documents that demand manual effort to decode

Zero Cross-Test Intelligence

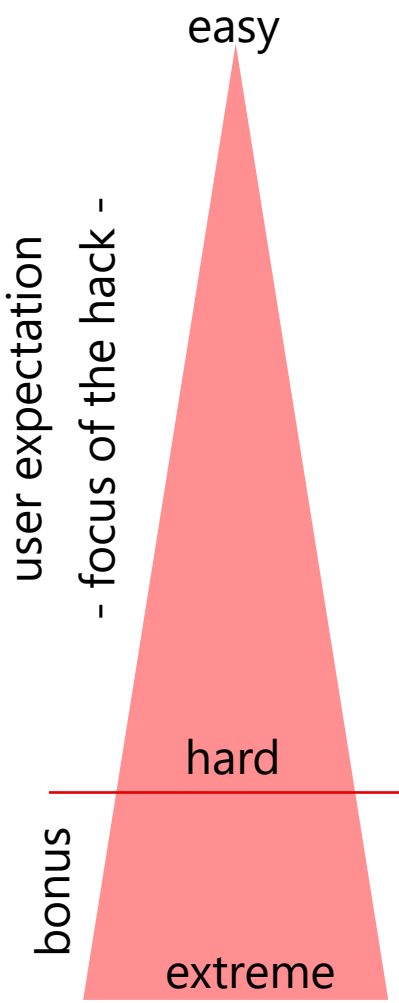
Each test stands alone — comparisons are painful or impossible

No Natural Query Layer

No intuitive way to ask: "*What's really going on in my data?*"

> **"The problem is not that the data is bad — it's that it's scattered across"**

Typical questions raised by our users

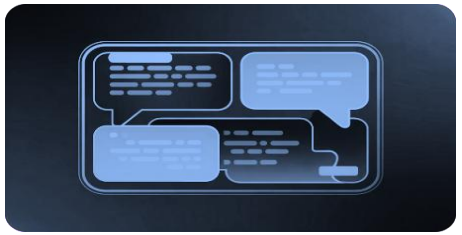


- **Data selection & queries**
  - Show me the compression test results we performed on 4th of May for Empire Industries for their Stardust material.
  - Summarize all available material properties for Fancyplast 42.
  - Show me all test we have done for customer Megaplant.
  - List all charpy impact tests performed by tester MasterOfDesaster.
- **Data comparison**
  - Compare material FancyPlast 42 and UltraPlast 99 regarding property tensile strength — are the differences statistically significant?
  - Do my Z05 and Z20 machines produce significantly different results?
  - Do my Ulm and Kennesaw sites deliver comparable quality
- **Data analytics**
  - Is there a degradation in the tensile strength of the material Hostacomp G2
  - Is there a trend that tensile strength is decreasing over the last 6 months in my local Plant?
  - Is there an indication that boundary 10 MPa tensile modulus of material FancyPlast 42 will be violated in the future?

- **Plausibility & Interpretation (Bonus)**
  - Is the measured value plausible and as expected to industry standard ISO 1234
  - Is the measured tensile strength of Fancyplast 42 within the range of my internal guidelines and limits
- **Complex hypotheses (Bonus)**
  - If I change parameter A, how does that influence property B?

What an Ideal Solution Looks Like?

A high-level, unconstrained vision of the future of materials intelligence



Natural Language Interface

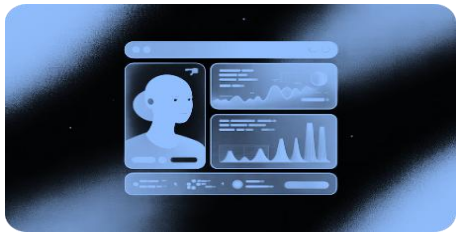
Engineers ask questions in plain English - no query language required

Context-Aware Intelligence

Understands follow-ups, retrieves relevant insights, and explains results

Cross-Dataset Reasoning

Compares tests, summarises reports, and visualises key metrics across tests



Smart Expert Assistant

Acts like a senior materials engineer - always available, infinitely patient

> **"Imagine talking to your data like you talk to a colleague"**

Your Vision

We're Not Here to Constrain You!

This hackathon is about your vision. We're deliberately leaving the solution space wide open

🎨 **No Prescribed UI** Build whatever interface feels right: chat, voice, visual, or something entirely new – working prototype would be nice!	⚙️ **No Stack Rules** Use any language, framework, or tooling you love — Python, TypeScript, Rust, anything
🤖 **Any LLM** Open AI, Claude, Gemini, open-source — your call entirely	📂 **Any Data Strategy** RAG, embeddings, fine-tuning, SQL agents — the architecture is yours to design

> **The only rule: "We want your ideas, not a prescribed solution"**

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What Good Looks Like?

No rigid requirements — just a north star to aim for

⚡ **Speed to Insight**
Dramatically reduce the time between "I have data" and "I understand it"

💬 **Practical Q&A**
Answer real engineering questions — failure modes, tolerances, anomalies

📊 **Compare & Summaries**
Surface comparisons and report summaries that would take hours manually

🧩 **Conversational Flow**
Handle follow-up questions naturally — no need to start over each time

> **"If your solution helps understand data faster - that's already a win"**

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Tech & Data

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Overlaying Structure

- Data is provided as a MongoDB
- Essential Collections:
  - Tests
    - testParameters define the test
    - E.g. customer, material, standard, etc.
  - Values
    - Stores the Results or the measurement data

We recommend using **MongoDB Compass** to get a feeling for the Data

When implementing a MCP (Model Context Protocol) Server, we recommend using **MCP inspector** to quickly test your MCPs

We recommend using **Python** with **fastMCP, pandas, numpy** to quickly get a suite up and running

Test Documents


```
_id: "{D1CB87C7-D89F-4583-9DA8-5372DC59F25A}"
clientAppType : "testXpert III"
state : "finishedOK"
▸ tags : Array (1)
version : "2.1772195387.0"
▸ valueColumns : Array (574)
hasMachineConfigurationInfo : false
testProgramId : "{EF33D0CB-B231-448e-937F-92876240DF2F}-ZugVorlage_V2.zp2"
testProgramVersion : "2.1772195387.0"
name : "01"
comment : null
▸ TestParametersFlat : Object
```


- **TestParamtersFlat** contain the Parameters of this particular Test
- **valueColumns** contains references to the values in the valueColumn Collection

Tests to Values


```
{
  "_id": "{D1CB87C7-D89F-4583-9DA8-5372DC59F25A}",
  "clientAppType": "testXpert III",
  "state": "finishedOK",
  "tags": [...],
  "version": "2.1772195387.0",
  "valueColumns": [
    {...},
    {
      "segments": [...],
      "unitTableId": "Zwick.Unittable.Displacement",
      "valueTableId": "{E4C21909-B178-4fdc-8662-A13B4C7FF756}-Zwick.Unittable.Displacement",
      "_id": "{E4C21909-B178-4fdc-8662-A13B4C7FF756}-Zwick.Unittable.Displacement_Value",
      "name": "Strain / Deformation"
    },
```


These are used to refence the value collection

Measurements can be identified by their UUID + Unit of Measurement.

- We already added a name field to most value entries to make getting started easier
- A list of UUIDs translations will also be provided

ValueColumn Documents


```
{
  "timestamp": {
    "$date": "2026-02-27T12:29:59.399Z"
  },
  "metadata": {
    "childId": "{E4C21909-B178-4fdc-8662-A13B4C7FF756}-Zwick.Unittable.Displacement.{E4C21909-B178-4fdc-8662-A13B4C7FF756}-Zwick.Un
    "fileId": "69a18e47467aa52ae03aaaa5",
    "filename": "%7BD1CB87C7-D89F-4583-9DA8-5372DC59F25A%7D-2.1772195387.0-%7BE4C21909-B178-4fdc-8662-A13B4C7FF756%7D-Zwick.Unittab
    "length": 359288,
    "refId": "{D1CB87C7-D89F-4583-9DA8-5372DC59F25A}",
    "rootVersion": "2.1772195387.0",
    "uploadDate": {
      "$date": "2026-02-27T12:29:59.399Z"
    }
  },
  "values": [
    0.00001000857524573803,
    0.00001006615161895752,
    0.00000983762834221 1247,
    0.00001023613847792 1485,
    0,
    -2.4464416128466838e-9,
```


Values are separated between _key and _value entries

Tip: ignore the _key entries as they don't usually contain valuable information

Metadata:

- refId: refers to the test ID
- childId: refers to the test.valuecolumn entry via the following schema:
  - `[test.valuecolumn[].valuetableId].[test.valucolumn[]._id]`

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Additional Notes

- To accurately get Measurements with timestamps, you will have to extract the Time measuremnts and actual other measurements of your choice.
- Measurements may have NOT been taken in a regular interval

Model Context Protocol (MCP)

- Provides a **standardized way for AI models to interact with tools and databases**.
- Enables **context-aware AI agents** that can retrieve information and execute tasks in software systems.

MCP – Tools & Resources

- **Tools**
  - Functions that the AI can **execute to perform actions**
  - Examples: database queries, API calls, file operations, running code
  - Allow AI models to **interact with external systems**
  - Similar to function calling in modern AI APIs

- **Resources**
  - **Structured data sources** that the AI can read from
  - Provide contextual information to the model
  - Examples: documents, databases, files, knowledge bases
  - Typically **read-only access for retrieving information**

MCP - Prompts

- **Prompts**
  - **Predefined instructions or templates** that guide AI behavior
  - Help standardize tasks and interactions
  - Examples: workflows, system prompts, task instructions
  - Ensure **consistent AI responses and workflows**

What you will find in the repository

- The pitch and deep dive slides
- Some examples of old-fashioned static PDF reports (one PDF per test)
- The MongoDB filled with Data
- README.md with additional Information

- Link:
  - https://github.com/adriank71/ZwickRoell-START-Hack-2026-

START HACK

Let’s make data talk

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